WORKED SOLUTION

STEP II 2007, Question 5

Pure • Iterated Functions

STEP II 2007, Question 5

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/07-s2-q5

The key idea

The shape $\dfrac{x+\sqrt{3}}{1-\sqrt{3}\,x}$ is the compound-angle formula for $\tan(A+B)$ with $\tan B = \sqrt{3}$, that is $B = \frac{\pi}{3}$. So f simply adds $\frac{\pi}{3}$ to an angle, and since tan has period π, three steps bring us back where we started. In part (iii) the same trick works with sin and $\frac{\pi}{6}$, but there is a catch: $\sqrt{1-t^2}$ is a cosine only when the angle lies in $[-\frac{\pi}{2}, \frac{\pi}{2}]$, and that changes the answer.

Solution

Part (i)

Put $y = f(x) = \dfrac{x+\sqrt{3}}{1-\sqrt{3}\,x}$. Then $f^2(x) = \dfrac{y+\sqrt{3}}{1-\sqrt{3}\,y}$, and multiplying top and bottom by $1-\sqrt{3}\,x$,

$$f^2(x) = \frac{(x+\sqrt{3})+\sqrt{3}(1-\sqrt{3}\,x)}{(1-\sqrt{3}\,x)-\sqrt{3}(x+\sqrt{3})} = \frac{2\sqrt{3}-2x}{-2-2\sqrt{3}\,x} = \frac{x-\sqrt{3}}{1+\sqrt{3}\,x}.$$

Now apply f once more, this time with $y = f^2(x)$, multiplying top and bottom by $1+\sqrt{3}\,x$:

$$f^3(x) = \frac{(x-\sqrt{3})+\sqrt{3}(1+\sqrt{3}\,x)}{(1+\sqrt{3}\,x)-\sqrt{3}(x-\sqrt{3})} = \frac{4x}{4} = x.$$

So f^3 is the identity. (The two excluded points are exactly what keeps every step defined: $f(x)$ needs $x \neq 1/\sqrt{3}$, $f^2(x)$ needs $x \neq -1/\sqrt{3}$, and f maps $\mathbb{R} \setminus \{\pm 1/\sqrt{3}\}$ onto itself, so we can keep going.)

Since $2007 = 3 \times 669$,

$$f^{2007}(x) = (f^3)^{669}(x) = x.$$

Part (ii)

Let $x = \tan\theta$. For any angle α with $\tan\alpha$ defined and $1 - \sqrt{3}\tan\alpha \neq 0$, the compound-angle formula with $\tan\frac{\pi}{3} = \sqrt{3}$ gives

$$f(\tan\alpha) = \frac{\tan\alpha + \tan\frac{\pi}{3}}{1 - \tan\alpha\tan\frac{\pi}{3}} = \tan(\alpha + \tfrac{\pi}{3}).$$

With $\alpha = \theta$ this is the case $n = 1$. If $f^k(x) = \tan(\theta + \frac{1}{3}k\pi)$, then applying the formula with $\alpha = \theta + \frac{1}{3}k\pi$,

$$f^{k+1}(x) = f(\tan(\theta + \tfrac{1}{3}k\pi)) = \tan(\theta + \tfrac{1}{3}(k+1)\pi),$$

so by induction $f^n(x) = \tan(\theta + \frac{1}{3}n\pi)$ for every positive integer n.

Nothing breaks along the way: the formula fails only if some angle $\theta + \frac{1}{3}k\pi$ is congruent to $\frac{\pi}{2}$ or $\frac{\pi}{6}$ modulo π, and that would force $\theta = \pm\frac{\pi}{6}$ or $\pm\frac{\pi}{2}$, that is, $x = \pm 1/\sqrt{3}$ or x infinite, which are excluded. This also explains part (i): adding $\frac{\pi}{3}$ three times adds π, the period of tan.

Part (iii)

Since $|t| \leq 1$ we can write $t = \sin\varphi$ with $\varphi = \arcsin t \in [-\frac{\pi}{2}, \frac{\pi}{2}]$.

? **Ask yourself** Is $\sqrt{1 - \sin^2\alpha}$ always equal to $\cos\alpha$?

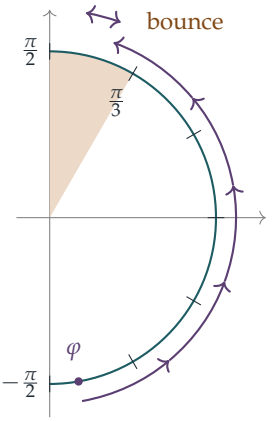
Only when $\cos\alpha \geq 0$. For an angle $\alpha \in [-\frac{\pi}{2}, \frac{\pi}{2}]$ it is fine, so

$$g(\sin\alpha) = \cos\tfrac{\pi}{6}\sin\alpha + \sin\tfrac{\pi}{6}\cos\alpha = \sin(\alpha + \tfrac{\pi}{6}). \qquad (*)$$

To apply $(*)$ again we must rewrite the answer as sin of an angle in $[-\frac{\pi}{2}, \frac{\pi}{2}]$. There are two cases.

- If $\alpha \leq \frac{\pi}{3}$, then $\alpha + \frac{\pi}{6} \leq \frac{\pi}{2}$ is already in range.
- If $\frac{\pi}{3} \leq \alpha \leq \frac{\pi}{2}$, then $\alpha + \frac{\pi}{6}$ may overshoot $\frac{\pi}{2}$, but $\sin(\alpha + \frac{\pi}{6}) = \sin(\pi - \alpha - \frac{\pi}{6}) = \sin(\frac{5\pi}{6} - \alpha)$, and $\frac{5\pi}{6} - \alpha \in [\frac{\pi}{3}, \frac{\pi}{2}]$.

(At $\alpha = \frac{\pi}{3}$ both give $\frac{\pi}{2}$.) So the in-range angle *climbs* by $\frac{\pi}{6}$ until it reaches $[\frac{\pi}{3}, \frac{\pi}{2}]$, and from then on it is *reflected* by $\alpha \mapsto \frac{5\pi}{6} - \alpha$, which is undone by doing it again: the angle bounces between two values.



Each application of g moves the angle $\frac{\pi}{6}$ round the right half of the unit circle; once in the shaded zone $[\frac{\pi}{3}, \frac{\pi}{2}]$ it flips back and forth.

Now let m be the number of climbs needed: the least integer $m \geq 0$ with $\varphi + \frac{m\pi}{6} \geq \frac{\pi}{3}$. Since $\varphi \geq -\frac{\pi}{2}$, $m \leq 5$, and in terms of t:

t	$[\frac{\sqrt{3}}{2}, 1]$	$[\frac{1}{2}, \frac{\sqrt{3}}{2})$	$[0, \frac{1}{2})$	$[-\frac{1}{2}, 0)$	$[-\frac{\sqrt{3}}{2}, -\frac{1}{2})$	$[-1, -\frac{\sqrt{3}}{2})$
m	0	1	2	3	4	5

Write $g^n(t) = \sin\varphi_n$ with $\varphi_n \in [-\frac{\pi}{2}, \frac{\pi}{2}]$ and $\varphi_0 = \varphi$. For $n \leq m$ every earlier angle was below $\frac{\pi}{3}$, so $\varphi_n = \varphi + \frac{n\pi}{6}$. Then $\varphi_m \in [\frac{\pi}{3}, \frac{\pi}{2}]$, so $\varphi_{m+1} = \frac{5\pi}{6} - \varphi_m$, $\varphi_{m+2} = \varphi_m$, and so on, alternating. Since $\sin\varphi_{m+1} = \sin(\varphi_m + \frac{\pi}{6}) = \sin(\varphi + \frac{(m+1)\pi}{6})$, we get, with $\varphi = \arcsin t$ and m from the table,

$$g^n(t) = \begin{cases} \sin(\varphi + \frac{n\pi}{6}), & n \leq m+1, \\ \sin(\varphi + \frac{m\pi}{6}), & n > m+1,\ n-m \text{ even}, \\ \sin(\varphi + \frac{(m+1)\pi}{6}), & n > m+1,\ n-m \text{ odd}. \end{cases}$$

In particular, for every t and every $n \geq 5$, $g^{n+2}(t) = g^n(t)$.

For example, $t = 0$ has $\varphi = 0$, $m = 2$, and the orbit is $0 \mapsto \frac{1}{2} \mapsto \frac{\sqrt{3}}{2} \mapsto 1 \mapsto \frac{\sqrt{3}}{2} \mapsto 1 \mapsto \cdots$.

Checks

- At $x = 1$: $f(1) = \frac{1+\sqrt{3}}{1-\sqrt{3}} = -2 - \sqrt{3}$, $f^2(1) = \frac{1-\sqrt{3}}{1+\sqrt{3}} = \sqrt{3} - 2$, and $f(\sqrt{3} - 2) = 1$, matching $\tan\frac{\pi}{4}$, $\tan\frac{7\pi}{12}$, $\tan\frac{11\pi}{12}$.
- Directly, $g(\frac{\sqrt{3}}{2}) = \frac{3}{4} + \frac{1}{4} = 1$ and $g(1) = \frac{\sqrt{3}}{2}$: the bounce is real. The naive guess $\sin\frac{5\pi}{6} = \frac{1}{2}$ for $g^5(0)$ is wrong; iterating gives 1.
- Numerically, iterating g nineteen times from 5000 random $t \in [-1, 1]$ agreed with the boxed formula to within 2×10^{-12}.

Common slips

- Answering (iii) with $\sin(\varphi + \frac{n\pi}{6})$ for all n. That forgets that $\sqrt{1-t^2} \geq 0$, so the angle can never pass $\frac{\pi}{2}$.
- Arithmetic slips in f^2: multiply numerator and denominator through by $1 - \sqrt{3}\,x$ before simplifying.

Beyond STEP

A map $x \mapsto \frac{ax+b}{cx+d}$ is a Möbius transformation, and composing two of them corresponds to multiplying their matrices $\left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right)$. Here f has matrix $2\left(\begin{smallmatrix} \cos\frac{\pi}{3} & \sin\frac{\pi}{3} \\ -\sin\frac{\pi}{3} & \cos\frac{\pi}{3} \end{smallmatrix}\right)$, a scaled rotation by $\frac{\pi}{3}$, whose cube is a multiple of the identity, which is why $f^3(x) = x$. Part (iii) is a small taste of dynamical systems: points that are *eventually periodic* rather than periodic. Needham's *Visual Complex Analysis* has a beautiful geometric treatment of Möbius transformations.

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